

C2 Input Verification

Validate C1→C2 handoff: classified chunks and motivation-filtered input

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June 30, 2026

Overview

Before building the C2 codebook or running evaluation, this notebook verifies the C1→C2 data handoff. Two targets are checked:

- `c1_classified_chunks`: C1 S2 classifications merged with chunk text and ground truth labels from `aligned_data`
- `c2_input_data`: Filtered to `pred_label == "FISCAL_MEASURE"` and `discusses_motivation == TRUE`

Eight tests check coverage, label completeness, text joins, C1 output fields, filter correctness, post-filter act coverage, class balance, and token distribution.

Part A: `c1_classified_chunks` (Precursor)

Test 1: Act Coverage

Every act in `aligned_data` should appear in `c1_classified_chunks`.

Test 1: PASS

Test 2: Label Completeness

Every row must have non-NA `motivation_category` and `exogenous_flag`.

Test 2: PASS

Test 3: Text Join Completeness

All rows should have non-NA, non-empty `text` after joining from `chunks`.

Test 3: PASS

Test 4: C1 Output Field Completeness

`pred_label` should be non-NA for all rows. Report the distribution of `discusses_motivation`, `discusses_timing`, and `discusses_magnitude`.

Table 1: C1 prediction distribution

Predicted Label	Chunks	Share
FISCAL_MEASURE	606	79.4%
NOT_FISCAL_MEASURE	157	20.6%

Table 2: Extra output field distribution (FISCAL_MEASURE chunks only)

Field	TRUE	FALSE	NA
motivation	509	97	0
timing	544	62	0
magnitude	539	67	0

Test 4: PASS

Part B: `c2_input_data` (Filtered)

Test 5: Filter Correctness

All rows must have `pred_label == "FISCAL_MEASURE"` and `discusses_motivation == TRUE`.

Test 5: PASS

Test 6: Act Coverage Post-Filter

How many aligned acts survive the motivation filter? Acts that lose all chunks cannot be evaluated by C2.

Test 6: PASS

Test 7: Motivation Class Balance

Distribution of `motivation_category` across C2 input chunks, compared to `aligned_data`.

Table 3: Motivation category: acts in `c2_input` vs `aligned_data`

Category	Exogenous?	<code>c2_input</code>	<code>aligned_data</code>
Long-run	Exogenous	15	15
Spending-driven	Endogenous	10	10
Deficit-driven	Exogenous	8	8
Countercyclical	Endogenous	6	6

Table 4: Chunk count per motivation category in c2_input

Category	Chunks
Long-run	212
Countercyclical	99
Deficit-driven	99
Spending-driven	99

Test 7: PASS

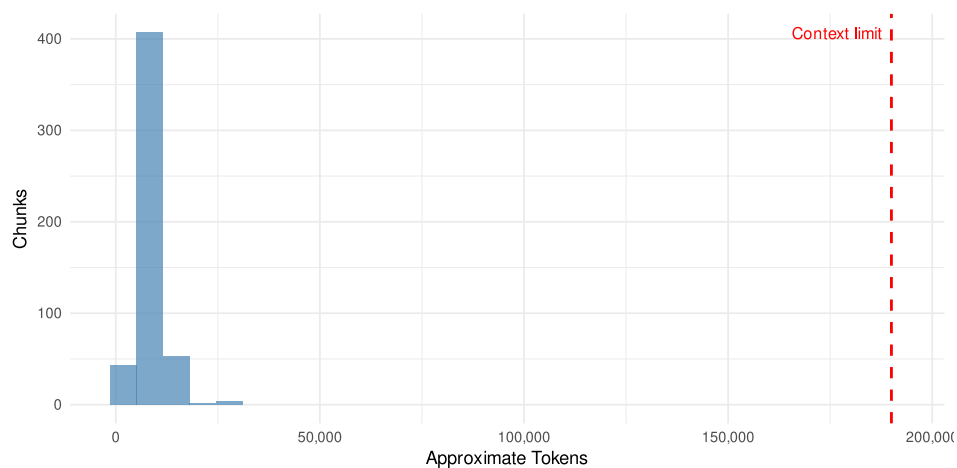
Test 8: Token Distribution

Per-chunk token distribution. Chunks were constructed to fit within the context window, so all should be well under 200K.

Table 5: Token distribution across C2 input chunks

Statistic	Tokens
min	1,794
p25	6,138
median	7,192
mean	7,880
p75	9,015
max	25,870

Figure 1: Approximate token count per chunk (C2 input)



Test 8: PASS

Per-Act Summary

Table 6: Per-act chunk summary: classified vs C2 input

Act	Year	Category	Classified	FISCAL_MEASURE	C2 Input
Tax Reform Act of 1969	1969	Counter-cyclical	27	25	24
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954	Spending-driven	24	21	20
Revenue Act of 1964	1964	Long-run	24	24	20
Revenue Act of 1978	1978	Long-run	24	21	20
Tax Adjustment Act of 1966	1966	Long-run	22	20	19
Tax Reform Act of 1986	1986	Long-run	20	19	19
Omnibus Budget Reconciliation Act of 1987	1987	Deficit-driven	22	19	18
Revenue Act of 1971	1971	Long-run	23	21	18
Revenue and Expenditure Control Act of 1968	1968	Counter-cyclical	22	20	18
Social Security Amendments of 1977	1977	Deficit-driven	21	19	17
Full Employment Act of 1946	1946	Counter-cyclical	23	21	17
Revenue Act of 1950	1950	Long-run	25	25	18
Revenue Act of 1954	1954	Long-run	25	25	18
Revenue Act of 1961	1961	Long-run	25	25	18
Revenue Act of 1962	1962	Long-run	25	25	18
Revenue Act of 1963	1963	Long-run	25	25	18
Revenue Act of 1964	1964	Long-run	25	25	18
Revenue Act of 1965	1965	Long-run	25	25	18
Revenue Act of 1966	1966	Long-run	25	25	18
Revenue Act of 1967	1967	Long-run	25	25	18
Revenue Act of 1968	1968	Long-run	25	25	18
Revenue Act of 1969	1969	Long-run	25	25	18
Revenue Act of 1970	1970	Long-run	25	25	18
Revenue Act of 1971	1971	Long-run	25	25	18
Revenue Act of 1972	1972	Long-run	25	25	18
Revenue Act of 1973	1973	Long-run	25	25	18
Revenue Act of 1974	1974	Long-run	25	25	18
Revenue Act of 1975	1975	Long-run	25	25	18
Revenue Act of 1976	1976	Long-run	25	25	18
Revenue Act of 1977	1977	Long-run	25	25	18
Revenue Act of 1978	1978	Long-run	25	25	18
Revenue Act of 1979	1979	Long-run	25	25	18
Revenue Act of 1980	1980	Long-run	25	25	18
Revenue Act of 1981	1981	Long-run	25	25	18
Revenue Act of 1982	1982	Long-run	25	25	18
Revenue Act of 1983	1983	Long-run	25	25	18
Revenue Act of 1984	1984	Long-run	25	25	18
Revenue Act of 1985	1985	Long-run	25	25	18
Revenue Act of 1986	1986	Long-run	25	25	18
Revenue Act of 1987	1987	Long-run	25	25	18
Revenue Act of 1988	1988	Long-run	25	25	18
Revenue Act of 1989	1989	Long-run	25	25	18
Revenue Act of 1990	1990	Long-run	25	25	18
Revenue Act of 1991	1991	Long-run	25	25	18
Revenue Act of 1992	1992	Long-run	25	25	18
Revenue Act of 1993	1993	Long-run	25	25	18
Revenue Act of 1994	1994	Long-run	25	25	18
Revenue Act of 1995	1995	Long-run	25	25	18
Revenue Act of 1996	1996	Long-run	25	25	18
Revenue Act of 1997	1997	Long-run	25	25	18
Revenue Act of 1998	1998	Long-run	25	25	18
Revenue Act of 1999	1999	Long-run	25	25	18
Revenue Act of 2000	2000	Long-run	25	25	18
Revenue Act of 2001	2001	Long-run	25	25	18
Revenue Act of 2002	2002	Long-run	25	25	18
Revenue Act of 2003	2003	Long-run	25	25	18
Revenue Act of 2004	2004	Long-run	25	25	18
Revenue Act of 2005	2005	Long-run	25	25	18
Revenue Act of 2006	2006	Long-run	25	25	18
Revenue Act of 2007	2007	Long-run	25	25	18
Revenue Act of 2008	2008	Long-run	25	25	18
Revenue Act of 2009	2009	Long-run	25	25	18
Revenue Act of 2010	2010	Long-run	25	25	18
Revenue Act of 2011	2011	Long-run	25	25	18
Revenue Act of 2012	2012	Long-run	25	25	18
Revenue Act of 2013	2013	Long-run	25	25	18
Revenue Act of 2014	2014	Long-run	25	25	18
Revenue Act of 2015	2015	Long-run	25	25	18
Revenue Act of 2016	2016	Long-run	25	25	18
Revenue Act of 2017	2017	Long-run	25	25	18
Revenue Act of 2018	2018	Long-run	25	25	18
Revenue Act of 2019	2019	Long-run	25	25	18
Revenue Act of 2020	2020	Long-run	25	25	18
Revenue Act of 2021	2021	Long-run	25	25	18
Revenue Act of 2022	2022	Long-run	25	25	18
Revenue Act of 2023	2023	Long-run	25	25	18
Revenue Act of 2024	2024	Long-run	25	25	18
Revenue Act of 2025	2025	Long-run	25	25	18

Summary Dashboard

Table 7: C2 input verification dashboard

Test	Result	Status
T1: Act coverage	39/39 acts present	PASS
T2: Label completeness	motivation: 0 NA, exogenous: 0 NA	PASS
T3: Text join completeness	0 NA, 0 empty (of 763 rows)	PASS
T4: C1 output fields	pred_label: 0 NA; discusses_motivation: 509 TRUE / 97 FALSE / 157 NA	PASS
T5: Filter correctness	Non-FISCAL_MEASURE: 0, discusses_motivation != TRUE: 0	PASS
T6: Act coverage post-filter	39/39 acts retained (0 lost)	PASS
T7: Motivation class balance	4/4 categories match aligned_data	PASS
T8: Token budget fit	Max: 25,870 tokens (limit: 190,000)	PASS

Overall: 8 PASS, 0 WARN, 0 FAIL out of 8 tests.

All clear: C1→C2 handoff validated. Data is ready for C2 codebook development.